

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

**Before the Honorable Doris Johnson Hines
Administrative Law Judge**

In the Matter of:

CERTAIN WEARABLE DEVICES

Investigation No. 337-TA-1478

COMPLAINANTS' RESPONSIVE CLAIM CONSTRUCTION BRIEF

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Respondents' Exhibits 1–7 were filed with Respondents' Opening Claim Construction Brief, Complainants' Exhibits 1–7 were filed with Complainants' Opening Claim Construction Brief, and Complainants' Exhibit 8 is being filed concurrently with Complainants' Responsive Claim Construction Brief.

Ex. No.	Description
Respondents' Exhibits	
1	Expert Declaration of Tajana S. Rosing, Ph.D.
2	Exhibit 3 to Complainants' Oura Ring 4 Infringement Claim Chart – U.S. Patent No. 10,231,675 (served on March 16, 2026)
3	The American Heritage Dictionary of the English Language, 5th ed. (2011)
4	Dictionary of Computer Vision and Image Processing (2005)
5	Dictionary of Information Security (2006)
6	Merriam-Webster's Collegiate Dictionary, 11th ed. (2012)
7	Newton's Telecom Dictionary, 27th ed. (2013)
Complainants' Exhibits	
1	U.S. Patent No. 10,231,675 (SAMS382-0003655)
2	U.S. Patent No. 10,642,359 (SAMS382-0003704)
3	Oura's Invalidity Contentions [excerpted to include only cited sections]
4	Newton's Telecom Dictionary 27th ed. (2013) (OURA1478_00002999)
5	Dictionary of Computer Vision and Image Processing (2005) (OURA1478_00002989)
6	Declaration of Darrin Young, Ph.D.
7	Curriculum Vitae of Darrin J. Young, Ph.D.
8	'359 Patent File History, Preliminary Amendment [excerpted and cited passages highlighted]

I. INTRODUCTION

Oura's brief confirms its perfunctory approach to claim construction. Rather than engage with the intrinsic record, Oura instead tries to leverage superficial ambiguities and typographical errors in the claim language to assert indefiniteness or entirely revamp the claim meaning to support non-infringement. But even a cursory review of the intrinsic evidence resolves the purported ambiguities and compels the constructions proposed by Samsung.

For the two terms of U.S. Patent No. 10,231,675 (the "'675 Patent") relating to the three recited time "periods" (i.e., the "first period," "second period," and "third period"), Oura feigns confusion about when each period starts or ends and the relationship between the periods. But any such confusion is readily dispelled by the claim language itself, as well as the specification, which provides ample guidance regarding the scope of the recited "periods." Eschewing the intrinsic evidence, Oura's leading arguments for indefiniteness are premised on Samsung's *infringement contentions*, which do not purport to define the metes and bounds of each term and are irrelevant to deciding indefiniteness.

Both parties acknowledge an obvious typographical error in the last term, "the biometric signal" in Claim 14 of U.S. Patent No. 10,642,359 ("'359 Patent"). Samsung's proposed construction corrects this to the similar phrase, "the biological signal," recited earlier in the claim, and is supported by the language of the claim itself, the specification, and other claims in the patent. Oura does not engage with these intrinsic sources. Instead, Oura proposes correcting "the biometric signal" to "a biometric signal" and defining "biometric" based entirely on extrinsic dictionary definitions. As a consequence, Oura's proposed construction adds new matter to the patent and the claims with no support in the specification or other intrinsic evidence.

Samsung's proposed constructions should, therefore, be adopted and Oura's contrary arguments rejected.

II. ANALYSIS OF DISPUTED CLAIM TERMS

- A. **Term No. 1: “acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body” (’675 Patent, Claim 13)**

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Within this phrase, “second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body” means: “second period of time which begins: (i) before the first period of time has ended when it is determined that the electronic device is worn, or (ii) immediately after the first period of time has ended when it is determined that the electronic device is worn”	Indefinite

As Samsung’s opening brief detailed, the scope of the “first period” and “second period” in Claim 13 and their timing relationship are evident from the claim language itself, and confirmed by the specification. As Claim 13 recites, in each recited “period,” the claimed “electronic device” uses a “first sensor” to measure the device’s “motion.” Compl. Op. Br. at 6-7 (citing Cl. 13, 19:47-49, 20:13-16, 20:64-21:8). The distinction between the different “periods” is *when* each period occurs and the *purpose* of the motion measurement in each period. *Id.* The “first period” is defined by measuring the device’s motion for the purpose of detecting whether the device is worn by the user. *Id.* The “second period” happens after wear detection in the “first period” and is defined by measuring the device’s motion for the different purpose of determining whether to activate a “second sensor” that measures “biometric information” of the user. Claim 13 also recites two scenarios for the relative timing of the two periods: the “second period” can either (i) start “shortly before” the “first period” ends (i.e., the two periods overlap), or (ii) follow “immediately after” the “first period” (i.e., the two periods do not overlap, but follow consecutively). Samsung’s construction accurately captures both possibilities.

Oura's brief attempts to sow confusion on two points: (1) how the "first" and "second" periods are identified and defined; and (2) what it means for the "second" period to be "shortly before and/or immediately after" the "first" period. Resp. Op. Br. at 9. On both points, Oura's confusion is readily resolved by the ample guidance provided by the specification.

1. The "first" and "second" periods are well-defined by their distinct timing and functions

As Oura acknowledges, the test for indefiniteness is whether "[t]he claims, *when read in light of the specification and the prosecution history* . . . provide objective boundaries for those of skill in the art." Resp. Op. Br. at 8-9 (citing *Interval Licensing LLC v. AOL, Inc.*, 766 F.3d 1364, 1371 (Fed. Cir. 2014)). Oura's leading argument, however, misdirects the ALJ's attention to Samsung's *infringement contentions* for purported support. Resp. Op. Br. at 9. Oura claims Samsung's "*allegations* are purely subjective" because the accused "first period of time is a period of time when the electronic ring is not being worn," while the "second period" is "when a user is wearing the ring and may end when the user wakes up." *Id.* at 10.

Oura's reliance on Samsung's contentions in an effort to show indefiniteness is misguided and legally incorrect. Samsung's infringement allegations do not purport to define the metes and bounds of the "first" and "second" periods. Thus, Oura's assertion that those "*allegations* are purely subjective," even if true, says nothing about whether the disputed claim language is subjective.¹ As a matter of law, Samsung's contentions are not relevant to assess definiteness or the metes and bounds of claim terms. Instead, as Oura acknowledges, the relevant source is the

¹ Samsung's contentions only identify when the two "periods" occur in the context of the accused Oura products; they do not purport to define the full scope of the '675 Patent claims. Further, even if Samsung's contentions were relevant to the definiteness inquiry, Samsung's allegations are far from subjective: (1) the "first period" corresponds to a time when the claimed "electronic ring" is not yet worn or detected as worn, and (2) the "second period" is when a user has not yet woken up. There is nothing subjective about whether a user is or is not wearing a device, or resting or not resting. See Resp. Op. Br. at 11.

intrinsic record, including the claims and specification. Resp. Op. Br. at 9 (citing *Interval Licensing*).

Here, as quoted below, Claim 13 itself defines (1) the “first period” as a period where “motion” is measured for the purpose of “determining whether an electronic device is worn,” as shown in the claim’s first step and parts of the second step (identified in red), and (2) the “second period” as a period where “motion” is also measured, but for a different purpose of deciding whether to “activat[e] at least one second sensor” based on that motion, in the second and third steps (identified in blue):

13. A method comprising:
determining whether an electronic device is worn on a part of a user's body using
at least one first sensor of the electronic device;
acquiring first information on a motion of the electronic device using at least a part
of the at least one first sensor for a second period of time shortly before and/or
immediately after a first period of time when it is determined that the electronic
device is worn on a part of the user's body;
activating at least one second sensor based on at least a part of the first information
on the motion of the electronic device;
...

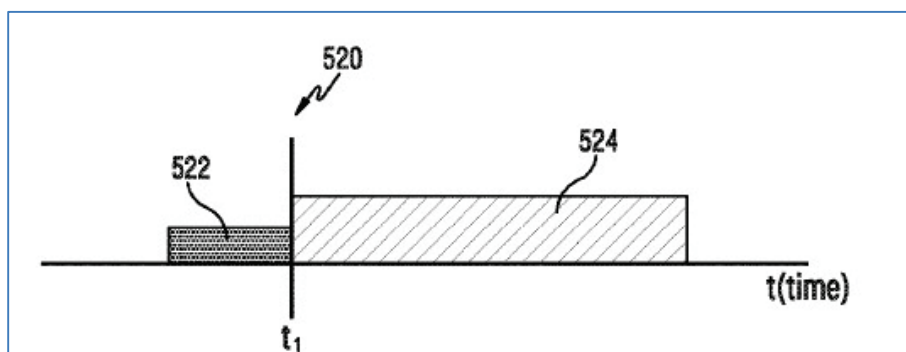
The specification confirms and elaborates on Claim 13’s recitation of the two periods as detailed in Samsung’s opening brief. *See* Compl. Op. Br. at 6-10. Indeed, the very passage in column 20, quoted in Oura’s brief and below, tracks the relevant steps of Claim 13 and confirms the scope and function of both periods:

[D]etermine whether the electronic device is worn on (or close to) a part of the user’s body using the at least one first sensor; acquire information on the motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time selected (or specified) passes from the time when the electronic device is worn on (or close to) a part of the user’s body; and activate (operate or trigger) the at least one second sensor based on at least a part of the information on the motion.

Compl. Op. Br., Ex. 1 (’675 Patent) at 20:63-21:8. In this passage, during the “first period,” described in the red text, motion measured by a “first sensor” is used to “determine whether the

electronic device is worn.” During the “second period,” described in the blue text, motion measured by the same “first sensor” is used for another purpose of determining whether to “activate (operate or trigger) the at least one second sensor.”

Aspects of the above passage in column 20 are further described in other passages and figures, such as Figure 5B’s timing diagram below, which illustrates two consecutive first and second timing intervals 522 and 524, in which a device’s motion is measured. *Id.* at 19:47-49, 20:13-16. In the first interval 522, which corresponds to the “first period” or part of that period, the device uses a first motion sensor to measure “motion” for one purpose, such as detecting whether the



device is worn. Compl. Op. Br., Ex. 6 (Young Decl.) ¶ 33. The specification elaborates on how wear detection works in this first interval or period, stating that “when the motion of the electronic device is detected through, for example,” a motion sensor (such as “an acceleration sensor, a terrestrial magnetism sensor, or a gyro sensor”), the device may then determine that it “has been worn on a user’s body.” Compl. Op. Br., Ex. 1 (’675 Patent) at 16:48-52. Then, in the second interval 524, which corresponds to the “second period,” the device continues to measure motion for a different purpose of deciding whether to activate another “sensor (e.g., biometric sensor) for measuring biometric information when the measured motion of the electronic device satisfies a selected (or specified) reference.” *Id.* at 20:25-29.

Thus, Claim 13, as further explained by the specification, provides clear criteria by which the “first” and “second” periods can be identified and specified. The “first period” can be

identified as a period where the device measures motion to determine if it is being worn. The “second period” can be identified as a period following wear detection, and where the device continues to measure motion to determine whether to activate another sensor for measuring biometric information.

2. “shortly before and/or immediately after” Is Also Well-Defined Based On The Specification

Oura also fails to fully consider the intrinsic record’s guidance when it contends that the phrase, “second period of time shortly before and/or immediately after a first period of time,” invokes terms of degree that lack defined, objective boundaries.

As an initial matter, Oura focuses on “shortly before,” but says little about why the scope of “immediately after” may be uncertain. “Immediately after” is readily understood from its plain meaning to mean that the “second period” follows the “first period,” without delay. As detailed in Samsung’s opening brief, “immediately after” describes one of two scenarios (with “shortly before” describing the other) for the timing relationship between the “first” and “second” periods, and it is depicted in Figure 5B (excerpted above) where time interval 524 immediately follows after interval 522. Compl. Op. Br. at 7-8 (citing Compl. Op. Br., Ex. 1 (’675 Patent) at 19:47-49, 20:13-16). Samsung’s construction captures this “immediately after” scenario, with the second part of its construction, “second period of time which begins . . . (ii) immediately after the first period of time has ended when it is determined that the electronic device is worn.”

As to “shortly before,” Oura’s brief laments that the specification leaves uncertain whether the “first period” could be “several days,” a “few hours,” or a “few minutes,” because under Samsung’s infringement contentions, the “first period” could be “any period of time when the electronic ring is not being worn.” Resp. Op. Br. at 12-13. But, as discussed, Samsung’s contentions do not purport to define the boundaries of the claim term “first period.” Rather, Claim

13 and the specification describe the “first period,” and both sources explain that the “first period” is not merely any time the device is unworn—rather, it is a specific period when the device is obtaining motion information to determine whether it is worn. That this “first period” is not specified as a fixed or set numeric duration in the claim does not render it indefinite, and Oura cites no authority suggesting otherwise.

Further, Oura contradicts itself by quoting column 18 of the specification, which gives exemplary durations of a period, such as “1 minute” or “10 seconds.” Resp. Op. Br. at 13 (quoting Compl. Op. Br., Ex. 1 (’675 Patent) at 18:19-45) (“the measurement module 506 may perform a measurement operation when a biometric information measurement period (e.g., about **1 minute**) arrives. ... the measurement module 506 may monitor a change in the motion for a predetermined period of time (e.g., about **10 seconds**).”). While Oura contends column 18 does not clearly relate to Claim 13 (*id.* at 13-14), other specification passages give example durations that clearly correspond to Claim 13. For example, in describing Figure 5B (excerpted above) that depicts a first interval 522 and a second interval 524, the specification explains that the first “time interval 522 [is] before the operation of the first timer ends (520) (e.g., for about **10 seconds** before the operation of the first timer ends).” Compl. Op. Br., Ex. 1 (’675 Patent) at 19:46-50. The second interval 524 that follows could be “e.g., **about 30 seconds**” and is used to “determine whether the motion of the electronic device is changed to satisfy the selected (or specified) reference” and therefore whether to “measure biometric information.” *Id.* at 20:1-4.

At any rate, Oura is fundamentally wrong in arguing that the objective boundaries for “shortly before” must be tied to numerical durations identified in the specification. While the specification gives exemplary time durations, the objective boundaries for “shortly before” do not arise from such examples, nor are such examples required to support definiteness. *See Enzo*

Biochem, Inc. v. Applera Corp., 599 F.3d 1325, 1335 (Fed. Cir. 2010) (finding term of degree definite even “without reference to a precise numerical measurement” in the construction).

Instead, the boundaries for “shortly before” arise from more objective criteria in the specification, such as triggering events that can cause the “second period” to start after the “first period” has started but before it ends. In other words, there may be a scenario where the device determines that it is worn before the “first period” for making that determination ends. This is exactly what Claim 13 means when it recites “shortly before . . . first period when it is determined that the electronic device is worn on a part of the user's body.”

Other parts of the specification, which Oura ignores, confirm that “shortly before” refers to a situation where a device is detected as worn before the end of the “first period” for making that determination, such that the “second period” can start before the “first period” ends. For example, the specification explains that the “second period of time” can begin “***shortly before*** and/or immediately ***after a first period of time selected (or specified) passes*** from the time when the electronic device is worn on (or close to) a part of the user’s body,” meaning the “second period” starts “before” the “first period” “passes” or ends. Compl. Op. Br., Ex. 1 (’675 Patent) at 21:3-5. This scenario, where the “second period” overlaps the “first period,” is also described in other passages stating that “the second period of time may be set to ***include the time point when the first period of time passes.***” *Id.* at 21:55-57, 42:21-23 (same).

These passages also refute Oura’s objection that Samsung’s construction wrongly removes the term of degree (“shortly”) and “broaden[s] the scope of the claim to allow for the second period of time to occur during the first period of time, as opposed to being a separate and distinct time period, as required by the claim.” Resp. Op. Br. at 15. There is nothing in the claim language that precludes the “first” and “second” periods from overlapping. To the contrary, as quoted above,

the specification repeatedly describes situations where the “second period” includes the “time point when the first period of time passes,” meaning the start of the “second period” overlaps with the end of the “first period.” Compl. Op. Br., Ex. 1 (’675 Patent) at 21:55-57, 42:21-23; *see also id.* at 21:52-54 (“at least one of the second period of time and the third period of time may be set ***to include the time point when the first period of time passes***”), 42:18-20 (same). “Shortly before” reflects this overlap scenario, where the trigger for starting the “second period” is “when the electronic device is worn,” which can occur “shortly before” the “first period” for detecting the wear condition ends. *Id.* at 21:3-5. As a result, the device can enter the “second period” without waiting for the “first period” for wear detection to end. The first portion of Samsung’s construction captures this “shortly before” scenario by construing that phrase to mean: “second period of time which begins: (i) before the first period of time has ended when it is determined that the electronic device is worn.”²

- B. Term No. 2: “acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and activating the at least one second sensor based on at least a part of the additional information” (’675 Patent, Claim 15)**

Samsung	Oura
Within this phrase, “third period of time” means: “period of time that immediately follows the second period of time and during which additional information regarding the motion of the electronic device is collected by at least a part of the at least one first sensor and used to determine whether to activate the at least one second sensor”	Indefinite

²There is also no relevance or merit to Oura’s contention that dependent Claim 15’s recitation of a “third period” that is “shorter than the first period,” somehow supports the indefiniteness of this disputed term in Claim 13 concerning the “first” and “second” periods. *See* Compl. Op. Br. at 14. To the extent Oura contends “shorter” is uncertain, it plainly means that the “third period” has a shorter duration or length of time than the “first period.” To the extent Oura contends the “third period” itself is indefinite, that is a separate issue addressed in the next section below.

As explained above and in Samsung’s opening brief, in each of the recited “periods,” the claimed “electronic device” measures motion, but the “periods” are distinguished by *when* they occur and the *purpose* the motion detection serves. In Claim 13, the “first period” uses the motion measurement for wear detection, and the “second period” uses the motion measurement to determine whether to activate or inactivate a “second sensor” for measuring “biometric information.” For the “third period” in Claim 15, the claim itself expressly describes its timing and function: it follows “immediately after the second period of time passes,” and its function is to “activat[e] the at least one second sensor” after that same “second sensor” was inactivated at the end of Claim 13.

In disputing the definiteness of the term, Oura ignores the clear guidance that Claim 15 itself provides and instead again leads with a misguided discussion of Samsung’s *infringement contentions*. Resp. Op. Br. at 16-17.³ As a matter of law, it is the intrinsic evidence—not Samsung’s contentions—that is the relevant source for defining the metes and bounds of terms and assessing their definiteness.

Here, as explained, Claim 15, on its face, is already clear about the metes and bounds of the “third period,” including:

1. When it starts: “immediately after the second period of time passes”;
2. The function or purpose served: determining whether to “activat[e] the at least one second sensor based on at least a part of the additional information,” after that “second sensor” was “inactivat[ed]” at the end of Claim 13, from which Claim 15 depends; and

³ Even assuming Samsung’s allegations could be relevant, defining the “third period” using the user’s wake or rest state, as determined by motion, is not subjective and does not leave unbounded or uncertain when the “third period” starts and ends. To the contrary, defining a “third period” by a user’s state of being awake or at rest, provides clear objective criteria, including that the period begins when the user is awake and ends when the user goes back to rest.

3. How long it lasts: “shorter than the first period of time.”

Given the express boundaries drawn in Claim 15 itself, there is no merit to Oura’s contention that “the only objective boundaries related to the third period of time are that it must be shorter than the first period of time.” Resp. Op. Br. at 17.

Moreover, beyond Claim 15, the specification confirms and elaborates on the boundaries for the “third period” set forth in the claim. As just one example, in the passage below, the specification explains that the “third period” (1) is “shorter than the first period,” (2) follows “immediately after the second period,” and (3) serves the purpose of acquiring “additional information” about “motion” to determine whether to “activate (operate or trigger) the at least one second sensor based on at least a part of the additional information,” corresponding to “motion,” depending on whether the motion “satisf[ies] a selected (or specified) condition”:

[A]cquire additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is ***[1] shorter than the first period of time, [2] immediately after the second period of time passes; and [3] activate (operate or trigger) the at least one second sensor based on at least a part of the additional information,*** without activating (operating or triggering) the second sensor when the information on the motion does not satisfy a selected (or specified) condition.

Compl. Op. Br., Ex. 1 (’675 Patent) at 21:42-51; *see id.* at 42:8-17 (similar).

Oura is also mistaken and relies on a false premise when it argues that, because the “first period” lacks objective boundaries, Claim 15’s requirement that the “third period” is shorter than the “first period” also lacks objective boundaries. Resp. Op. Br. at 17. As explained for the prior term, the “first period” has objective boundaries: it is a time period for determining whether the claimed “electronic device” is being worn, using motion information. Thus, the “third period,” which is for using motion to decide whether to reactivate a second sensor, must have a shorter duration than that “first period” for wear determination.

Oura’s indefiniteness proposal should therefore be rejected. To the extent any construction is required, “third period” within the disputed term should be construed as Samsung proposes: “period of time that immediately follows the second period of time and during which additional information regarding the motion of the electronic device is collected by at least a part of the at least one first sensor and used to determine whether to activate the at least one second sensor.”

C. Term No. 3: “[the] biometric signal” (’359 Patent, Claim 14)

Samsung	Oura
Correcting “the biometric signal” to be “the biological signal” and applying the plain and ordinary meaning (i.e., a signal that is associated with biological information of the user).	Correcting “the biometric signal” to be “a biometric signal” and applying the plain and ordinary meaning (i.e., a signal relating to the identification of an individual based on physical characteristics, such as fingerprints, finger lengths, facial features, retinal patterns, voice recognition, and weight).

The parties agree that the recitation of “the biometric signal” in Claim 14 contains an obvious typographical error that should be corrected. Compl. Op. Br. at 14; Resp. Op. Br. at 18. As Samsung’s opening brief explains, the claim language itself, the specification, and other claims in the ’359 Patent, all support the proper correction to read “the biological signal,” consistent with the antecedent recitation of “a biological signal” earlier in the claim. Compl. Op. Br. at 14-15.

Oura pays no heed to these intrinsic sources and instead asserts that Samsung’s correction improperly “change[s] the term and its meaning.” But it is Oura, not Samsung, that seeks to change the meaning of the term to manufacture a non-infringement argument based on a concept nowhere even discussed in the intrinsic record, namely the “identification of an individual.” Resp. Op. Br. at 18-19. Oura’s attempt to wholesale change the claimed invention should be rejected.

First, well-settled caselaw permits exactly the type of correction Samsung proposes. As Oura acknowledges, courts may “correct ‘obvious minor typographical and clerical errors in patents’” so long as “(1) the correction is not subject to reasonable debate based on consideration

of the claim language and the specification and (2) the prosecution history does not suggest a different interpretation of the claims.” *Pavo Sols. LLC v. Kingston Tech. Co., Inc.*, 35 F.4th 1367, 1373 (Fed. Cir. 2022) (cited in Resp. Op. Br. at 18).

In *Canatex Completion Sols., Inc. v. Wellmatics, LLC*, the Federal Circuit applied this standard on similar facts to reverse a district court that refused to correct a claim term reciting “the connection profile of the second part” to read “the connection profile of the first part.” 159 F.4th 39, 47-49 (Fed. Cir. 2025). In *Canatex*, the claim recited “a first part comprising an external connection profile” but later referred to “the connection profile of the second part”—a phrase that lacked any express antecedent. *Id.* at 42-43. The Federal Circuit concluded that a skilled artisan would “immediately see that, as written, there is an error in the claim” because the definite article “the” required an antecedent to “the second part” that did not exist. *Id.* at 47. The Federal Circuit then held that “the connection profile of the first part” was “the only reasonable correction” based on the claim language, the specification, and the figures. *Id.* at 48-49.

The court in *VTT Technical Research Centre of Finland Ltd. v. SiTime Corp.* similarly corrected a claim to resolve a missing antecedent basis. No. 4:19-cv-1174-YGR, 2020 WL 4193374, at *9 (N.D. Cal. July 21, 2020). There, a dependent claim 30 referred to “the temperature range of 100°C”—using the definite article “the”—but its base claim 29 did not include the required antecedent of a “temperature range of 100°C.” *Id.* The court found the error of a missing antecedent in claim 29 obvious from the face of the patent because every other reference to temperature drift in the claims and specification included a temperature range of 100°C. *Id.* The court corrected the error by appending an antecedent phrase “over a temperature range . . . that extends at least 100°C” into claim 29, holding that the correction was “not subject to reasonable debate” based on the claim language, the specification, and the prosecution history. *Id.* at *9-10.

Claim 14 of the '359 Patent presents an even more obvious typographical and antecedent basis error that can and should be judicially corrected. Although Claim 14 introduces “a biological signal” and “a biological sensor,” it later refers to “the biometric signal”—a phrase with no express antecedent but with a clear implicit antecedent in the earlier reference to “a biological signal”:

obtain a movement signal of an object external to the wearable device using the motion sensor;

obtain *a biological signal* of the object using *the biological sensor*;

determine a select period, of the movement signal, that meets a specified condition; determine, as a valid signal, at least a portion of *the biometric signal* that is determined to correspond to the determined select period; and

perform a specified function using the valid signal.

Following the Federal Circuit’s reasoning in *Canatex*, the article “the” before “biometric signal” must be a reference back to one of the two previously recited signals in Claim 14—“a movement signal” and “a biological signal.” *See also VTT*, 2020 WL 4193374, at *9-10 (fixing a missing antecedent by adding missing antecedent claim language consistent with the specification). Because “biometric signal” is etymologically closer to “biological signal” than to “movement signal,” and because the function of the “biometric signal” in the claim—being validated based on a “select period” of the “movement signal”—mirrors the function of the “biological signal” in the specification and other patent claims (e.g., Claim 7), rather than the “movement signal,” the only reasonable correction is “biological signal.” *See Compl. Op. Br.* at 14-17 (explaining how the “biological signal” is recited or described in the specification at 3:52-64, 5:14-17, 10:9-22, 10:63-11:2, 11:45-54, Fig. 4, and in Claim 7).

Second, it is Oura—not Samsung—who attempts to completely change the meaning of the claim with its proposal to change “the biometric signal” to “a biometric signal.” *Canatex* rejected a similar attempt by defendants. There, the defendants proposed corrections for the lack of express

antecedent basis, including changing “the” to “a” (as Oura suggests here) to correct a lack of antecedent. 159 F.4th at 48-49. The Federal Circuit rejected each proposal as “illogical” as it “would alter the scope and meaning of the claims significantly from what is clear from the claim language as a whole and is described in the specification.” *Id.* In that case, changing “the” to “a” would have completely changed the meaning of the claims by adding new subject matter because “[n]othing in the figures or their descriptions shows a ‘connection profile’ in the second part” and that “the figures clearly show ‘connection profile 16’ as being of first part 12.” *Id.* at 47-49.

The same problem arises here with Oura’s proposal to change “the biometric signal” to “a biometric signal,” and then defining “a biometric signal” as a signal used for “identification” of users. Oura’s proposal injects a wholly new concept—individual identification based on “fingerprints, finger lengths, facial features, retinal patterns, voice recognition, and weight”—that appears nowhere in the specification. The ’359 Patent has nothing to do with identifying users for security purposes. Other than the one typographical error in Claim 14, the ’359 Patent never uses the term “biometric” or “biometric signal” anywhere, nor does it reference any signals corresponding to fingerprints, facial features, voices, or the like for identifying individuals.

The ’359 Patent describes only “biological signal[s] or physiological signal[s]” for measuring “biological information” such as “muscle contraction and muscle relaxation.” Compl. Op. Br., Ex. 2 (’359 Patent) at 3:52-64, 5:14-17. Reflecting the intrinsic record’s focus on “biological information” and signals, Samsung proposes to correct “biometric signal” to “biological signal.” This involves a minor change to a word, “biological,” that shares the same root (“bio-”) and that already appears in the very same claim. The correction adds no new subject matter; rather, it aligns the disputed term with the antecedent phrase “a biological signal” that Claim 14 already recites, and it also aligns with the specification and every other claim in the

patent. *See* Compl. Op. Br. at 14-17. Thus, just as in *Canatex*, Samsung’s “simple, logical change” should be adopted. 159 F.4th at 49. Oura’s contrary proposal to import an entirely foreign concept into the patent is “not [a] reasonably debatable alternative[]” and should therefore be rejected. *Id.*

Third, the prosecution history does not support Oura’s interpretation, as Oura contends. Oura posits that, during prosecution, Samsung amended certain claims but left the term “biometric signal” un-amended during prosecution. Resp. Op. Br. at 19-20. But all this shows is that the erroneous use of “biometric signal” was already present in Claim 14, when it was first introduced in a preliminary amendment as Claim 15, and was never corrected during prosecution—a circumstance fully consistent with an overlooked typographical error. Ex. 8 (’359 Patent Preliminary Amendment) at 4. The fact that the applicant did not catch the error during prosecution does not demonstrate any intent to claim a “biometric signal” distinct from the “biological signal.” Oura identifies no prosecution history statement, amendment, or argument that affirmatively gives “biometric signal” a meaning distinct from “biological signal.”

Finally, Oura attempts to bury in a footnote the sources of its proposed construction of “biometric signal.” Resp. Op. Br. at 20 n.6. Those sources consist of five dictionary definitions of “biometric,” “biometrics,” or “biometric device.” *See id.* (citing Exs. 3-7). Conspicuously absent is any basis in the specification or other intrinsic source for Oura’s proposed definition, as the intrinsic sources instead only describe “biological signals” associated with a user’s “biological information.” *E.g.*, Compl. Op. Br., Ex. 2 (’359 Patent) at 3:52-64, 5:14-17. “[A]ny definition found in or ascertained by a reading of the intrinsic evidence may not be contradicted by any meaning found in dictionaries or technical treatises.” *Dow Chem. Co. v. Sumitomo Chem. Co.*, 257 F.3d 1364, 1373 (Fed. Cir. 2001). Thus, Oura’s wholesale reliance on extrinsic dictionary definitions to contradict the clear teachings of the intrinsic record is improper.

Based on that intrinsic record—including Claim 14 itself, the specification, and other claims in the '359 Patent—Oura's attempt to use “biometric signal” to revamp the claimed invention should be rejected. Instead, “biometric signal” should be corrected to “biological signal,” which is “a signal associated with biological information of the user.”

III. CONCLUSION

For the reasons discussed above and in Samsung's opening brief, Samsung respectfully requests that its constructions be adopted for the disputed terms in the '675 and '359 Patents.

Dated: May 18, 2026

Respectfully submitted,

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CERTIFICATE OF WORD COUNT

The undersigned hereby certifies that the foregoing Responsive Claim Construction Brief complies with the word limitations of Ground Rule 7.2. Excluding the items listed in Ground Rule 1.7 as not included in the word count, and using the word count of the word-processing system used to prepare the paper (Microsoft Word), Complainants' Responsive Claim Construction Brief contains 5,365 words, which is under the 8,000 word limit.

/s/ D. Sean Trainor

D. Sean Trainor

EXHIBIT 8

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re the Application of:

Chang Mok CHOI

Application No. 16/267,752

Art Unit: 2694

Confirmation No. 9126

Filed: February 5, 2019

Examiner: Gustavo D. Polo

For: WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF

RESPONSE PURSUANT TO 37 C.F.R. § 1.111

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

This is in response to the Non-Final Office Action mailed June 21, 2019 having a period for response set to expire on September 23, 2019.

The following amendments and remarks are respectfully submitted. Reconsideration of the claims is respectfully requested.

Amendments to the claims are reflected in the listing of the claims that begins on page 2 of this Paper.

Remarks begin on page 2 of this Paper.

IN THE CLAIMS

This listing of the claims replaces all prior versions and listings of the claims in this application.

The text of all pending claims (including any withdrawn claims) is set forth below. Canceled and not entered claims are indicated with claim number and status only. The claims as listed below show added text with underlining and deleted text with ~~strikethrough~~ or in [[double brackets]]. The status of each claim is listed with one of (Original), (Currently amended), (Canceled), (Withdrawn), (Previously presented), (New), and (Not entered).

Please **AMEND** claims 1, 2 and 4-20; **CANCEL** claim 6; and **ADD** claims 21-35 in accordance with the following:

1. (Currently amended) A wearable device comprising:

a motion sensor;

a biological sensor; and

a controller configured to:

obtain a movement signal of an object using the motion sensor, and a biological signal of the object using the biological sensor, ~~and~~

determine, based on an analysis of the movement signal, whether to perform a specified function of the wearable device using the biological signal, and

perform ~~a~~ the specified function using ~~at least a~~ select portion of the biological signal obtained during a ~~specified select period, that of~~ the movement signal, determined to corresponds to a specified condition.

2. (Currently amended) The wearable device of claim 1,

wherein the ~~controller is configured to perform, in response to specified condition~~ comprises the movement signal being determined to be less than or equal to a first threshold value, the wearable device using the biological signal.

3. (Original) The wearable device of claim 2, wherein the controller is configured to determine the first threshold value based on a magnitude of the biological signal.

4. (Currently amended) The wearable device of claim 1, wherein the controller is configured to determine the biological signal to be invalid, irrespective of the movement signal, in response to the magnitude of the biological signal not satisfying a second threshold value.

5. (Currently amended) The wearable device of claim 1, wherein, in response to a determination that the biological signal is valid, the controller is configured to operate in an active mode and to control an external device based on the biological signal.

6. (Canceled)

7. (Currently amended) The wearable device of claim 1, further comprising a transmission unit, wherein the controller is configured to transmit an instruction associated with an external electronic device to the external electronic device using the transmission unit as at least part of the specified function.

8. (Currently amended) A wearable device comprising:

a motion sensor;

a biological sensor; and

a controller configured to:

~~obtain, with respect to an object external to the wearable device, a movement signal of the~~
~~an object external to the wearable device using the motion sensor; and~~

~~obtain a biological signal of the object using the biological sensor;~~

~~identify a period, that of the movement signal, the movement signal is determined to be~~
~~smaller-less than a specified condition; and~~

~~perform a specified function using at least a select portion of the biological signal~~
~~corresponding to the period.~~

9. (Currently amended) The wearable device of claim 8, wherein the biological sensor comprises an optical sensor configured to obtain an optical signal ~~related~~ corresponding to the biological signal.

10. (Currently amended) The wearable device of claim 8, wherein the controller is configured to adjust, prior to the identifying of the period, the specified condition based at least in part on a magnitude of the biological signal ~~prior to the identifying of the period.~~

11. (Currently amended) The wearable device of claim 8, wherein the controller is configured to:

~~in response to a magnitude of the biological signal being determined to be within a first specified range, determine the biological signal as to be a valid signal if that a magnitude of the biological signal falls into a first specified range; and~~

~~in response to the magnitude of the biological signal being determined to be within a second specified range, determine the biological signal as to be an invalid signal if that a magnitude of the biological signal falls into a second specified range.~~

12. (Currently amended) The wearable device of claim 8, wherein, for the performing of the specified function, the controller is configured to generate an instruction using the ~~at least select~~ portion of the biological signal ~~as at least part of the performing~~.

13. (Currently amended) The wearable device of claim 12, wherein, for the performing of the specified function, the controller is configured to control an external electronic device based at least in part on the instruction as at least part of the specified function.

14. (Currently amended) The wearable device of claim 13, further comprising a transmission unit, wherein the controller is configured to transmit the instruction to the external electronic device using the transmission unit.

15. (Currently amended) A wearable device comprising:
a motion sensor;
a biological sensor; and
a controller configured to:

~~obtain, with respect to an object external to the wearable device, a movement signal of the an object external to the wearable device using the motion sensor; and~~

~~obtain a biological signal of the object using the biological sensor;~~

~~determine a select period of the movement signal that meets a specified condition;~~

~~determine, as a valid signal, at least a portion of the biometric signal that is determined to correspond to the determined ~~corresponding to a select period that the movement is smaller than a~~ specified condition as a valid signal; and~~

~~perform a specified function using the valid signal.~~

16. (Currently amended) The wearable device of claim 15, wherein the biological sensor comprises an optical sensor configured to obtain an optical signal ~~related to~~ the biological signal.

17. (Currently amended) The wearable device of claim 15, wherein the controller is configured to adjust the specified condition based at least in part on a magnitude of the biological signal prior to the ~~identifying-determining~~ of the ~~select~~ period.

18. (Currently amended) The wearable device of claim 15, wherein, ~~for the performing of the specified function,~~ the controller is configured to generate an instruction using ~~the~~ at least a portion of the valid signal ~~as at least part of the performing.~~

19. (Currently amended) The wearable device of claim 18, wherein, ~~for the performing of the specified function,~~ the controller is configured to control an external electronic device based at least in part on the instruction as at least part of the specified function.

20. (Currently amended) The wearable device of claim 19, further comprising a transmission unit, wherein the controller is configured to transmit the instruction to the external device using the transmission unit.

21. (New) The wearable device of claim 1, wherein
the controller is further configured to validate the biological signal, and
for the performing of the specified function, the controller is configured to perform the specified function using a portion of the validated biological signal.

22. (New) A wearable device system including one or more wearable devices, the wearable device system comprising:

a motion sensor of the one or more wearable devices;

a biological sensor of the one or more wearable devices; and

a controller of the one or more wearable devices, the controller configured to:

obtain, based on motion information from the motion sensor, a movement signal of an object;

obtain, based on biological information from the biological sensor, a biological signal of the object;

determine whether a period of the movement signal exists that meets a specified motion type indicating condition;

REMARKS

The Office is thanked for the review and consideration of the pending application. The Non-Final Office Action mailed June 21, 2019 ("Office Action") has been received and its contents reviewed. Reconsideration of the pending application is respectfully requested in view of the amendments and the following remarks.

In accordance with the foregoing, claims 1, 2 and 4-20 have been amended; claim 6 has been canceled; and claims 21-35 have been added. Accordingly, claims 1-35 are pending, with claims 1, 8, 15, and 22 being independent claims.

The amendments address typographical errors, improve readability, and clarify the nature of aspects of the claims. No new matter is presented in the amendments and newly added claims. Applicant has amended claims solely for the purpose of furthering the prosecution of a disclosed embodiment, and not in response to the merits of the outstanding rejections. Applicant reserves the right to pursue other claims, including broader claims or embodiments, in a continuation application.

Claim Rejections Pursuant to 35 U.S.C. § 103

Claims 1-20 stand rejected pursuant to 35 U.S.C. § 103 as allegedly being unpatentable over U.S. Patent App. Pub. No. 2010/0145171 to Park et al. ("Park") in view of U.S. Patent App. Pub. No. 2012/0157789 to Kangas et al. ("Kangas"). These rejections are respectively traversed.

Independent Claims 1, 8 & 15

In rejecting independent claim 1, the Office sets forth that the combination of Park and Kangas discloses all the claimed features. Applicant respectfully disagrees and respectfully submits that neither Park, Kangas, nor any combination thereof discloses or suggests all the claimed features of independent claim 1.

- 1. Perform the specified function using a select portion of the biological signal obtained during a select period, of the movement signal, determined to correspond to a specified condition*

As a non-limiting example, neither Park, Kangas, nor any combination thereof discloses "perform the specified function using a select portion of the biological signal obtained during a select period, of the movement signal, determined to correspond to a specified condition," as recited in independent claim 1.

For example, in its rejection of independent claim 1, the Office asserts:

Park is silent on perform a specified function using at least a portion of the biological signal obtained during a specified period that the movement corresponds to a specified condition. Kangas teaches perform a specified function using at least a portion of the biological signal obtained during a specified period [¶ 96 *bio-signal information which is stored in the bio-signal profile 25 may comprise data which has been collected, over an extended period of time*] that the movement corresponds to a specified condition [¶ 110 *by collecting data over a period of time and storing it in a bio-signal profile 25 the outputs provided are personal to the user of the apparatus and may be optimized to make the apparatus as convenient as possible for the user*].

Office Action at 2, 3.

As shown, it appears that the Office asserts that Kangas's collecting and/or storing of bio-signal information discloses performing a specified function using at least a portion of a biological signal.

However, while Kangas discloses collecting and storing bio-signal information to generate or update a bio-signal profile for a user, the bio-signal information is not collected "using a select portion of the biological signal obtained during a select period, of the movement signal, determined to correspond to a specified condition," as recited in independent claim 1.

For example, Kangas merely discloses "The bio-signal information which is stored in the bio-signal profile 25 may comprise data which has been collected, over an extended period of time, while the user uses the apparatus 1. The data may be collected by using the method illustrated in FIG. 3. In some embodiments of the invention the method illustrated in FIG. 3 may be repeated at regular or predetermined intervals." Kangas at ¶ 0096.

As shown, Kangas merely collects the bio-signal information at predetermined intervals, and this bio-signal information collected at predetermined intervals fails to disclose "a select portion of the biological signal obtained during a select period, of the movement signal, determined to correspond to a specified condition," as recited in independent claim 1.

Moreover, while Kangas discloses that "A bio-mechanical signal may comprise the user of the apparatus 1 moving a part of their body," Kangas at ¶ 0066, Kangas merely discloses that the bio-signal information may be of any one of "a bio-electrical signal, a bio-mechanical signal, an aural signal, a chemical signal or an optical signal." Kangas at ¶ 0063.

Accordingly, an embodiment of Kangas collects and analyzes one of the aforementioned types of bio-signals, and Kangas fails to disclose performing a specified function using a select portion of a bio-electrical, aural, a chemical, or optical signal obtained during a select period, of a bio-mechanical signal, determined to correspond to a specified condition.

Accordingly, Kangas fails to disclose “using a select portion of the biological signal obtained during a select period, of the movement signal, determined to correspond to a specified condition,” as recited in independent claim 1.

II. Determine, based on an analysis of the movement signal, whether to perform a specified function of the wearable device using the biological signal

As a non-limiting example, neither Park, Kangas, nor any combination thereof discloses “determine, based on an analysis of the movement signal, whether to perform a specified function of the wearable device using the biological signal,” as recited in independent claim 1.

As shown, features of previous claim 6 have been incorporated into independent claim 1.

In its rejection of claim 6, the Office appears to rely on Kangas with respect to measured biological signal of Kangas without consideration of the claimed relationship of “the movement” the Office Action has interpreted of Park, and the required relationship between the interpreted “movement” of Park and the pulse wave signal of Park.

For example, independent claim 1 now sets forth “determine, based on an analysis of the movement signal, whether to perform a specified function of the wearable device using the biological signal.”

Here, while Kangas may perform different functions based on different determined biological signals, there is no suggestion that whether those functions are performed have any dependence on the acceleration signal of the motion signal of Park or any similarly utilized motion signal.

The motion signal of Park is particularly derived from a motion sensor designed for collecting biosignals particularly related, and using a motion sensor particularly positioned, to/for removing motion artifacts or noise from the measured pulse wave signal.

This motion sensor would not be so similarly placed, or a movement signal determined therefrom, for the purpose of Park and for any related purpose of the different biosignal dependent function implementations of Kangas. Thus, neither Park nor Kangas, nor any combination of the same, could or would disclose “determine, based on an analysis of the movement signal, whether to perform a specified function of the wearable device using the biological signal,” as recited in independent claim 1.

Park fails to cure the aforementioned deficiencies in Kangas with respect to independent claim 1. It is further respectfully submitted that the combination of Park and Kangas also fails to disclose each and every claimed feature of each of independent claims 8 and 15.

III. Obviousness

In addition, it is respectfully submitted that the Office Action’s rejection rationale for independent claims 1, 8, and 15 is improper.

For emphases, in full, the Office Action sets forth:

1. Park discloses a wearable device [Fig. 1] comprising: a motion sensor [Fig. 2, 150]; a biological sensor [130]; and a controller [110] configured to: obtain a movement of an object using the motion sensor [¶ 50], and a biological signal of the object using the biological sensor [¶ 47]. Park is silent on perform a specified function using at least a portion of the biological signal obtained during a specified period that the movement corresponds to a specified condition. Kangas teaches perform a specified function using at least a portion of the biological signal obtained during a specified period [¶ 96 bio-signal information which is stored in the bio-signal profile 25 may comprise data which has been collected, over an extended period of time] that the movement corresponds to a specified condition [¶ 110 by collecting data over a period of time and storing it in a bio-signal profile 25 the outputs provided are personal to the user of the apparatus and may be optimized to make the apparatus as convenient as possible for the user]. Therefore it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify Park with Kangas as required by this claim, since such a modification improves the performance of the device.

Here, the Office Action has only argued that Kangas sets forth a feature that Park does not already have, and that the addition of that feature to Park would “improve the performance of the device of Park.”

It is respectfully submitted that a prima facie case of obviousness is not met when the rejection rationale does not include an explanation of why Park would add the particular feature of Kangas, other than the addition would provide the additional feature of the Kangas and ‘improve performance’ of the Park device. The Office Action has not explained why the performance of Park would be increased, or why one skilled in the art would look to the particular additional feature of Kangas and find a reason for modifying Park to include that feature outside of the disclosure of the present application.

Park uses a measured motion for removing motion artifacts from a sensed biosignal.

For example, paragraph [0003] of Park sets forth:

The present invention relates to an apparatus for measuring a motion noise robust pulse wave and a method thereof, and more particularly, to an apparatus for measuring a motion noise robust pulse wave at the back of a wrist, i.e., that part of the wrist which extends into the back of the hand, through an adaptive signal processing technique by estimating a motion component using an acceleration sensor and a method thereof.

Further, independent claim 1 of Park further explains how noise is estimated based on the acceleration signal:

a signal preprocessor that estimates a motion noise model corresponding to the motion of the examinee by predicting the motion of the examinee on the basis of an acceleration signal detected by the acceleration sensor and removes motion noise for the pulse wave signal from the motion noise model.

Thus, Park presents a device directed to producing a more noiseless pulse wave signal by estimating the noise in the measured pulse wave signal based on the acceleration signal.

For example, see paragraph [0006] discussing that the pulse wave signal may be used to measuring a health index of a user. As another example, paragraph [0032] of Park sets forth:

In addition, with an increase in concern about health, the present invention may be used as a health measurement means for adults or old men and an emergency precursor situation detection function and when contents for stress, concentration, sleep quality, etc. through HRV analysis are provided to a host device, the present invention may be used as more various purposes, that is, for a game, education, wellbeing contents and in addition, when the wireless protocol such as Zigbee is used, since data of a plurality of wearable apparatuses in groups such as a hospital, a senior center, and a silver town can be collected and processed, the present invention may be used as a basic apparatus in a ubiquitous-based health care field.

Thus, there is no suggestion in Park to use the measured pulse wave signal for anything other than for more accurate health related contents or information.

Accordingly, the Office is proposing to modify Park to include an additional function, not needed or necessary with the invention of Park, to use the noiseless pulse wave signal of Park for controlling operations of the device. This operation would be independent of the noise removing operation of Park based on the motion sensor of Park, as it must occur after the noise removing operations of Park to achieve the benefit of Park

It is respectfully submitted that the addition of controlling a device based on the noiseless pulse signal of Park does not improve the operation of Park, but only adds a function that Park did not suggest, need, or desire.

For example, in the PTAB decision of *Kranos v Riddel* (Case IPR2016-1649, Feb 7, 2018), where the PTA explained:

The Supreme Court made clear that we apply “an expansive and flexible approach” to the question of obviousness. *KSR*, 550 U.S. at 415. Whether a patent claiming the combination of prior art elements would have been obvious is determined by whether the improvement is more than the predictable use of prior art elements according to their established functions. *Id.* at 417. To reach this conclusion, however, it is not enough to show merely that the prior art includes separate references covering each separate limitation in a challenged claim. *Unigene Labs., Inc. v. Apotex, Inc.*, 655 F.3d 1352, 1360 (Fed. Cir. 2011). Rather, obviousness additionally requires that a person of ordinary skill at the time of the invention “would have selected and combined those prior art elements in the normal course of research and development to yield the claimed invention.” *Id.*

As another example, see *Arendi v Apple*, (Fed. Cir. 2015-2073, August 10, 2016) setting forth:

[T]here are at least three caveats to note in applying “common sense” in an obviousness analysis. First, common sense is typically invoked to provide a known motivation to combine, not to supply a missing claim limitation. In *DyStar*, a pre-*KSR* case, we held a patent

obvious where “all claim limitations [were] found in a number of prior art references,” 464 F.3d at 1360, and a person of ordinary skill in the art would have combined the references to achieve a “cheaper, faster, and more convenient” process. *Id.* at 1371 (quoting *Sandt Tech., Ltd. v. Resco Metal & Plastics Corp.*, 264 F.3d 1344, 1355 (Fed. Cir. 2001)). We

stated that: “an implicit motivation to combine exists not only when a suggestion may be gleaned from the prior art as a whole, but when the ‘improvement’ is technology-independent and the combination of references results in a product or process that is more desirable.” DyStar, 464 F.3d at 1368. Similarly, in Randall, we invoked common sense to vacate the Board’s determination of non-obviousness where “the Board failed to consider a wealth of well-documented knowledge that is highly material to evaluating the motivation to combine references.” Randall, 733 F.3d at 1356. There, the Board’s decision to ignore the existence of a “prevalent, perhaps even predominant method” of achieving the limitation at issue was prejudicial error. *Id.* at 1363.

Second, in Perfect Web, the only case Appellees identifies in which common sense was invoked to supply a limitation that was admittedly missing from the prior art, the limitation in question was unusually simple and the technology particularly straightforward. “The patented invention involves comparing the number of successfully delivered e-mail messages in a delivery against a predetermined desired quantity, and if the delivery does not reach the desired quantity, repeating the process of selecting and e-mailing a group of customers until the desired number of delivered messages has been achieved.” Perfect Web, 587 F.3d at 1326. There, the missing claim limitation—step D of steps A-D—was nothing more than an instruction to repeat steps A, B, and C until a particular quantity of email was sent in accordance with the claim. By contrast, the missing search at issue here “plays a major role in the subject matter claimed” and “affects much more than step (i).” Appellant’s Br. 24. That is—if the search in step (i) is missing, then “the claims would be almost void of content” because the premise of the patent is to use information in a first program to find related information in a second program. *Id.* at 25. Thus, the facts in Perfect Web are distinguishable from the case at bar and ought to be treated as the exception, rather than the rule.

Third, our cases repeatedly warn that references to “*common sense*”—*whether to supply a motivation to combine or a missing limitation—cannot be used as a wholesale substitute for reasoned analysis and evidentiary support, especially when dealing with a limitation missing from the prior art references specified.* Indeed, we stated that although there is no problem with using common sense “without any specific hint or suggestion in a particular reference,” the Board’s “utter failure to explain the ‘common knowledge and common sense’ on which it relied” is problematic. DyStar, 464 F.3d at 1366 (explaining our reasoning in *In re Lee*, 277 F.3d 1338, 1341, 1344 (Fed. Cir. 2002)). See also *In re Zurko*, 258 F.3d 1379, 1383, 1385 (Fed. Cir. 2001) (reversing Board where it adopted examiner’s unsupported assertion that claim limitation missing from cited references was “basic knowledge” and it “would have been nothing more than good common sense” to combine the references).

For example, in *In re Lee*, the Board adopted the examiner’s statements during prosecution that combining two prior art references to achieve the claimed invention—a method of automatically displaying the functions of a video display device and demonstrating how to select and adjust the functions—would have been obvious to a person of ordinary skill. The examiner had stated that the combination would have been obvious “since the demonstration mode is just a programmable feature which can be used in many different devices for providing automatic introduction by adding the proper programming software,” and that “another motivation would be that the automatic demonstration mode is user friendly and it functions as a tutorial.” *In re Lee*, 277 F.3d 1338, 1341 (Fed. Cir. 2002). We vacated the Board’s decision,

holding that “[c]onclusory statements such as those here provided do not fulfill the agency’s obligation” to explain all material facts relating to a motivation to combine. *Id.* at 1344.

...
Based on this prior precedent, we conclude that while “common sense” can be invoked, even potentially to supply a limitation missing from the prior art, it must still be supported by evidence and a reasoned explanation. In cases in which “common sense” is used to supply a missing limitation, as distinct from a motivation to combine, moreover, our search for a reasoned basis for resort to common sense must be searching. And, this is particularly true where the missing limitation goes to the heart of an invention.

See also, *In re: Marcel Van Os* (Fed. Cir. 2015-1975, January 3, 2017), and *In re: Nuvasive* (Fed. Cir. 2015-1670, December 7, 2016)

Here, as noted above the Office Action’s suggested motivation of improving Park is not related to the invention of purpose of Park, and thus would appear to be merely a conclusion that if the additional functions of Kangas were added after the removing of the noise of the pulse wave signal, then the combination would be an improvement over Park, which would appear to be a conclusion that the addition of the additional features of Kangas would have been common sense for creating a larger combination what includes the features of Park in whole and then the identified features of Kangas. However, as noted above, it is respectfully submitted that this conclusion is improper.

Based at least on the foregoing, the rejections of independent claims 1, 8, and 15 pursuant to 35 U.S.C § 103 based on Park and Kangas are deficient, and Applicant respectfully requests the rejections be withdrawn.

Dependent Claims

Claims 2-7, 9-14, and 16-21 are dependent upon independent claim 1, 8, or 15 and incorporate all the respective features therein. Accordingly, the 35 U.S.C. § 103 rejections of the dependent claims based on Park and Kangas are deficient for at least the same rationale as applied to respective independent claim 1, 8, or 15 above, and Applicant respectfully requests the rejections be withdrawn.

In addition, it is respectfully submitted that the rejections of dependent claims 2-6, 10, 12-13, and 17-19 are improper. As noted above, a substantial evidence rejection rationale must be provided for each rejection.

Rather, the rejections of dependent claims 2-6, 10, 12-13, and 17-19 merely identify additional features of Kangas, without an explanation why the further modification of the combination of Park and Kangas would have been obvious, including an explanation of why one of ordinary skill in the art would make the additional modification.

It is respectfully submitted that these rejections are improper, and respectfully submitted that any next Office Action must be non-Final and clearly present the missing motivation for the Office's proposed further modifications of the combination applied to the independent claims, as it is respectfully submitted that the rejections of the independent claims do not state that or suggest that the entirety of the invention and disclosure of Kangas are modified into the invention of Park, and further provide no substantial evidence reasons for such a substantial change in Park.

Accordingly, withdrawal of the rejections of all dependent claims is further respectfully requested.

Conclusion

There being no further outstanding objections or rejections, it is submitted that the application is in condition for allowance. An early action to that effect is courteously solicited.

In the event this paper is not being timely filed, Applicant respectfully petitions for an appropriate extension of time. Any fees for such an extension together with any additional fees only associated with the processing of this Response and any other documents filed concurrently with this Response may be charged to Counsel's Deposit Account 50-5113. Finally, if there are any formal matters remaining after this response, the Office is requested to telephone the undersigned to attend to these matters.

Respectfully submitted,

Date: September 23, 2019

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CERTIFICATE OF SERVICE

Investigation No. 337-TA-1478

The undersigned certifies that on May 18, 2026, true and correct copies of the foregoing document were served upon the following parties as indicated below:

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